OSPF - Day 27
Part - II

Dec - 28

OSPF Cost

- OSPF's metric is called cost.
- It is automatically calculated based on the bandwidth (speed) of the interface.
- It is calculated by dividing a reference bandwidth value by interface's bandwidth.

- The default reference bandwidth is 100 Mbps.

Reference : 100 mbps / Interface : 10 mbps = cost of 10.
Reference : 100 mbps / Interface : 100 mbps = cost of 1
Reference : 100 mbps / Interface : 1000 mbps = cost of 1?
Reference : 100 mbps / Interface : 10000 mbps = cost of 1 ??

- All values less than 1 will be converted to 1.
- Therefore FastEthernet, Gigabit Ethernet, 10 GiG Ethernet, etc. are equal & all have cost of 1 by default.

We'll use network topology of figure 14.
(OSPF का कोई भी कमांड जैसे ip ospf जैसे commonly like sho ip ospf brief)
Lets see cost of R3 F2/0 interface.

R3 # show ip ospf interface.
Cost: 1

- We can & should change the reference bandwith with this command.

R1(config-router) # auto-cost reference-bandwidth mbps.

R3(config-router) # auto-cost reference-bandwidth 100000

The command is entered in megabits per second (default is 100)

- 100000/100 = cost of 1000 for FastEthernet
  100000/1000 = cost of 100 for Gig Ethernet.

- You should configure a reference bandwidth greater than the fastest link in your network (to allow for future upgrades)

- We should configure the same reference bandwidth on all OSPF routers in the network.

So we've now set (suppose) same OSPF ref bandwidth of 100 Gbps on all routers?

OSPF Cost

- The OSPF cost to a destination is the total cost of the outgoing/exit interfaces.
- For eg: R1's cost to 192.168.4.0/24 is:
  100 (R1 G0/0) + 100 (R2 G0/0) + 100 (R4 G1/0)
  = 300

- Loopback interfaces have a cost of 1.
- What is R1's cost to reach 2.2.2.2 (R2's loopback 0 int)?
  It exits R1's G0/0 & R2's loopback 0. Now it doesn't actually exit any physical interface to reach the virtual loopback interface. but the cost of 1 is added to metric.
  ie 100 (R1 G1/0) + 1 (R2 L0) = 101

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Manually configuring OSPF cost.

R1(config-if) # ip ospf cost 70000

This cost takes priority over auto-calculated cost.

OSPF cost contd...

One more option to change the OSPF cost of an interface is to change the bandwidth of the interface with the bandwidth command.

- The formula to calculate OSPF cost is $\frac{\text{reference bandwith}}{\text{interface bandwidth}}$.
- Although the bandwidth ~~manages~~ matches the interface speed by default, changing the interfaces bandwidth doesn't actually change the speed at which the interface operates.
- The bandwidth is just a value that is used to calculate OSPF cost, EIGRP metric etc.
- To change the speed at which int operates, use speed command.

If we change bandwidth of gigabit Ethernet int, to 100 mbps, it will still operate at 1Gbps. However, for purpose of OSPF's cost calculation, the bandwidth of 100 mbps will be used.

- Because the bandwidth value is used in other calculation, its not recommended to change this value to alter the interface's OSPF cost.

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- It is recommended that we change the reference bandwidth, and then use the ip ospf cost command to change the cost of individual interfaces if we want.

Changing bandwidth of an interface.

To change bandwidth of interface use the below command:

R1(config)# bandwidth bandwidth_in_kbps

OSPF cost continued.....

Three ways to modify the OSPF cost:

1) Change the reference bandwidth:
R1(config-router)# auto-cost reference-bandwidth megabits_per_second

2) Manual configuration
R1(config-if)# ip ospf cost <cost>

3) Change the interface bandwidth
R1(config-if)# bandwidth kilobits-per-second

Sshow ip ospf int brief - to see info of each ospf enabled int}

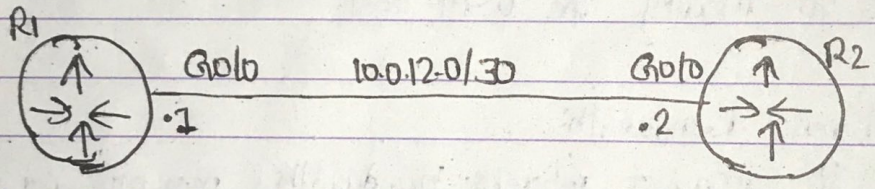
OSPF neighbors

- Making sure that routers successfully become OSPF neighbors is the main task in configuring and trouble shooting ospf.
- Once routers become neighbors, they automatically do the work of sharing network information, calculating routes etc.

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- When OSPF is activated on an interface, the router starts sending OSPF hello mssg out of interface at regular intervals (determined by hello timer). These are used to introduce the router to potential OSPF neighbors.

- The default hello timer is 10 seconds on Ethernet connection.
- Hello mssg are multicast to 224.0.0.5 (multicast add. for all OSPF routers.
- OSPF mssg are encapsulated in an IP header, with a value of 89 in the protocols field



For OSP routers to become neighbors, they need to go through different stages.

Down → Init → 2-way → Exstart → Exchange → Loading → Full

Steps to be OSPF neighbors

① ~~Init State~~ Down State.

② When R2 receives the Hello Packet, it will add an entry for R1 to its OSPF neighbor table.

In R2's neighbor table, the relationship with R1 is now on "init" state

a) OSPF is activated on R1's Gi0/0 interface.
b) It sends an OSPF hello message to 224.0.0.5
c) It doesn't know about any OSP neighbors yet, so the current neighbor state is down.

2) Init State.

- When R2 receives hello packet, it will add an entry for R1 to its OSPF neighbor state. In R2's neighbor table, the relationship with R1 is now in Init state.
  Init state = Hello packet received, but own router ID isn't in Hello packet.

3) 2 way State.

- R2 will send a Hello packet containing the RID of both routers.
- R1 will insert R2 into its OSPF neighbor table in the 2 way state.
- R1 will send another Hello message, this time containing R2's RID. Now both routers are in the 2-way state.

- The 2-way state means the router has received a Hello packet with its own RID in it.

- If both routers reach the 2-way state, it means that all of the conditions have been met for them to be OSPF neighbors. They're now ready to share LSAs to build a common LSDB.

- In some network types, a DR (Designated Router) and BDR (Backup Designated Router) will be elected at this point.
  (We will talk about OSPF network types and DR/BDR elections in Day 28)

4) Ex-Start State.

- 2 routers will now prepare to exchange information about their LSDB.

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- Before that, they have to chose which one will start the exchange. So they decide which one'll be master & which one'll be slave.
  (This is diffrnt than DR and BDR)
  The Master-Slave relationship is only needed for initial exchang of LSDB information.

- They do this in exstart state
- The router with higher RID'll became MASTER & initiat exchange. The router with the lower RID will become the slave.
- To decide Master & Slave, they exchange DBD (Data base description packets)

5a) # Exchange State.

- In the Exchange state, the routers exchange DBDs which contain a list of LSAs in their LSDB
- These DBDs don't include detailed information about the LSAs just basic information.
- The routers compare the information in the DBD they receive to the information in their own LSDB to determine which LSAs they must receive from their neighbor

6) Loading State

- In the Loading State, routers send Link State Request (LSR) messages to request that their neighbors send them any LSAs they don't have

- LSAs are sent in Link State Update (LSU) messages
- The routers send LSAck messages to acknowledge that they have received the LSAs

7) ~~In the~~ Full State

- In the Full state, the routers have a full OSPF adjacency & identical LSDBs.
- They continue to send and listen for Hello packets (every 10 sec by defaut) to maintain neighbor adjacency.
- Every time a Hello Packet is received, the 'Dead' timer (40 secs by default) is reset.
- If the Dead timer counts down to 0 and no Hello message is received, the neighbor is removed.
- The routers will continue to share LSAs as the network changes to make sure each router has a complete and accurate map of network. (LSDB)

OSPF message types.

Type	Name	Purpose
1	Hello	. Neighbor discovery & maintenance.
2	Database description (DBD).	Summary of the LSDB of the router. Used to check if LSDB of each router is same.
3	Link-State Request (LSR).	Requests specific LSAs from the neighbor.
4	Link-State Update (LSU).	Sends the specific LSAs to neighbor.
5	Link-State Acknowledgement (LSAck)	Used to acknowledge that the router received a message.

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Some OSPF commands

show ip ospf neighbor
show ip ospf interface
show ip ospf interface g0/0

{We can enable ospf directly on an interface, without using Network Command. Here we assumed ospf isn't enabled on R1 g0/0.}

R1(config)# int g0/0
R1(config-if)# ip ospf 1 area 0

{We can configure ALL interfaces as OSPF passive interface, then remove on those int which we don't want to be passive}

R1(config-router)# passive-interface default.
R1(config-router)# no passive-interface int-id.